# 4

# Motion in a Plane

## Motion in a Plane with Constant Acceleration

1. The position vector of a moving body at any instant of time is given as $\vec{r} = (5t^2\hat{i} - 5t\,\hat{j})$ m. The magnitude and direction of velocity at $t = 2s$ is, **[24 Jan, 2025 (Shift-II)]**
   (a) $5\sqrt{15}$ m/s, making an angle of $\tan^{-1} 4$ with –ve Y axis
   (b) $5\sqrt{17}$ m/s, making an angle of $\tan^{-1} 4$ with +ve X axis
   (c) $5\sqrt{15}$ m/s, making an angle of $\tan^{-1} 4$ with +ve X axis
   (d) $5\sqrt{17}$ m/s, making an angle of $\tan^{-1} 4$ with –ve Y axis

2. Position of an ant ($S$ in metres) moving in Y-Z plane is given by $S = 2t^2\hat{j} + 5\hat{k}$ (where $t$ is in second). The magnitude and direction of velocity of the ant at $t = 1$ s will be: **[27 Jan, 2024 (Shift-I)]**
   (a) 16 m/s in $y$-direction (b) 4 m/s in $x$-direction
   (c) 9 m/s in $z$-direction (d) 4 m/s in $y$-direction

3. A particle starts from origin at $t = 0$ with a velocity $5\hat{i}$ m/s and moves in $x$-$y$ plane under action of a force which produces a constant acceleration of $(3\hat{i} + 2\hat{j})$m/s². If the x-coordinate of the particle at that instant is 84 m, then the speed of the particle at this time is $\sqrt{\alpha}$ m/s. The value of $\alpha$ is ______. **[27 Jan, 2024 (Shift-I)]**

4. The co-ordinates of a particle moving in $x$-$y$ plane are given by:
   $x = 2 + 4t, \quad y = 3t + 8t^2$.
   The motion of the particle is: **[04 April, 2024 (Shift-I)]**
   (a) non-uniformly accelerated.
   (b) uniformly accelerated having motion along a straight line.
   (c) uniform motion along a straight line.
   (d) uniformly accelerated having motion along a parabolic path.

5. A force on an object of mass 100 g is $(10\hat{i} + 5\hat{j})$N. The position of that object at $t = 2$s is $(a\hat{i} + b\hat{j})$m after starting from rest. The value of $a/b$ will be ___________. **[25 June, 2022 (Shift-I)]**

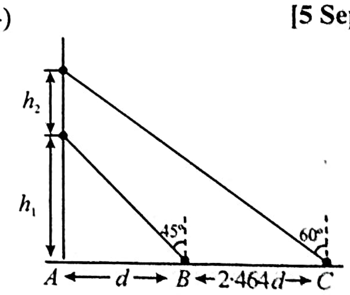
6. At time $t = 0$ a particle starts travelling from a height $7\hat{z}$ cm in a plane keeping $z$ coordinate constant. At any instant of time it's position along the $\hat{x}$ and $\hat{y}$ directions are defined as $3t$ and $5t^3$ respectively. At $t = 1s$ acceleration of the particle will be **[28 July, 2022 (Shift-II)]**
   (a) $-30\hat{y}$ (b) $30\hat{y}$ (c) $3\hat{x} + 15\hat{y}$ (d) $3\hat{x} + 15\hat{y} + 7\hat{z}$

7. A particle is moving along the $x$-axis with its coordinate with time '$t$' given by $x(t) = 10 + 8t - 3t^2$. Another particle is moving along the $y$-axis with its coordinate as a function of time given by $y(t) = 5 - 8t^3$. At $t = 1s$, the speed of the second particle as measured in the frame of the first particle is given as $\sqrt{v}$. Then $v$(in m/s) is **[8 Jan, 2020 (Shift-I)]**

8. A particle starts from the origin at $t = 0$ with an initial velocity of $3.0\,\hat{i}$ m/s and moves in the $x$-$y$ plane with a constant acceleration $(6.0\hat{i} + 4.0\hat{j})$ m/s². The $x$-coordinate of the particle at the instant when its $y$-coordinate is 32 m is $D$ meters. The value of $D$ is **[9 Jan, 2020 (Shift-II)]**
   (a) 50 (b) 40 (c) 32 (d) 60

9. Starting from the origin at time $t = 0$, with initial velocity $5\hat{j}$ m/s, a particle moves in the $x$-$y$ plane with a constant acceleration of $(10\hat{i} + 4\hat{j})$ m/s². At time $t$, its coordinates are (20 m, $y_0$ m). The values of $t$ and $y_0$ are, respectively **[4 Sep, 2020 (Shift-I)]**
   (a) 4 s and 52 m (b) 2 s and 24 m
   (c) 5 s and 25 m (d) 2 s and 18 m

10. A balloon is moving up in air vertically above a point $A$ on the ground. When it is at a height $h_1$, a girl standing at a distance $d$ (point $B$) from $A$ (see figure) sees it at an angle 45° with respect to the vertical. When the balloon climbs up a further height $h_2$, it is seen at an angle 60° with respect to the vertical if the girl moves further by a distance 2.464 $d$ (point $C$). Then the height $h_2$ is (given $\tan 30° = 0.5774$) **[5 Sep, 2020 (Shift-I)]**




   (a) $d$ (b) $0.732\,d$ (c) $1.464\,d$ (d) $0.464\,d$

11. A particle is moving with a velocity $\vec{v} = K(y\hat{i} + x\hat{j})$, where K is a constant. The general equation for its path is: **[9 Jan, 2019 (Shift-I)]**
   (a) $y = x^2$ + constant (b) $y^2 = x$ + constant
   (c) $y^2 = x^2$ + constant (d) $xy$ = constant

12. A particle moves from the point $(2.0\hat{i} + 4.0\hat{j})$m, at $t = 0$ with an initial velocity $(5.0\hat{i} + 4.0\hat{j})\text{ms}^{-1}$. It is acted upon by a constant force which produces a constant acceleration $(4.0\hat{i} + 4.0\hat{j})\text{ms}^{-2}$. What is the distance of the particle from the origin at time 2s? **[11 Jan, 2019 (Shift-II)]**
   (a) 15 m (b) $20\sqrt{2}$ m (c) 5 m (d) $10\sqrt{2}$ m

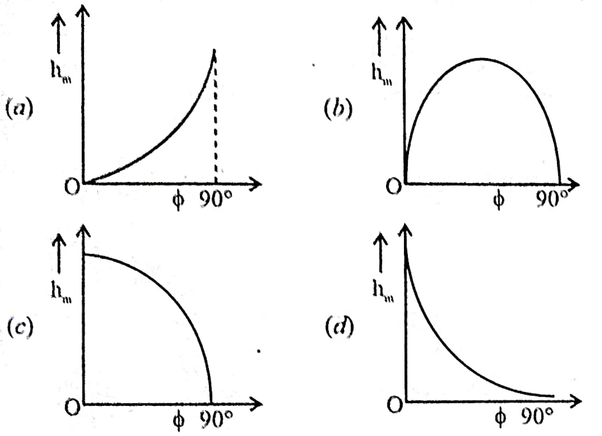
13. The position co-ordinates of a particle moving in a 3-D coordinates system is given by
   $x = a\cos\omega t$
   $y = a\sin\omega t$
   and $z = a\omega t$
   The speed of the particle is: **[9 Jan, 2019 (Shift-II)]**
   (a) $\sqrt{2}\,a\omega$ (b) $a\omega$ (c) $\sqrt{3}\,a\omega$ (d) $2a\omega$

14. The position vector of a particle changes with time according to the relation $\vec{r}(t) = 15t^2\hat{i} + (4 - 20t^2)\hat{j}$. What is the magnitude of the acceleration at $t = 1$? [9 April, 2019 (Shift-II)]
(a) $40\ m/s^2$ (b) $100\ m/s^2$ (c) $25\ m/s^2$ (d) $50\ m/s^2$

## "Projectile Motion: Ground to Ground Projection"

15. A particle is projected with velocity $u$ so that its horizontal range is three times the maximum height attained by it. The horizontal range of the projectile is given as $\frac{nu^2}{25g}$, where value of $n$ is :
(Given 'g' is the acceleration due to gravity.)
[03 April, 2025 (Shift-II)]
(a) 6 (b) 18 (c) 12 (d) 24

16. The angle of projection of a particle is measured from the vertical axis as $\phi$ and the maximum height reached by the particle is $h_m$. Here $h_m$ as function of $\phi$ can be presented as [03 April, 2025 (Shift-I)]




17. Two balls with same mass and initial velocity, are projected at different angles in such a way that maximum height reached by first ball is 8 times higher than that of the second ball. $T_1$ and $T_2$ are the total flying times of first and second ball, respectively, then the ratio of $T_1$ and $T_2$ is : [08 April, 2025 (Shift-II)]
(a) $2\sqrt{2}:1$ (b) $2:1$ (c) $\sqrt{2}:1$ (d) $4:1$

18. Two projectiles are fired from ground with same initial speeds from same point at angles $(45° + \alpha)$ and $(45° - \alpha)$ with horizontal direction. The ratio of their times of flights is [07 April, 2025 (Shift-I)]
(1) 1 (2) $\frac{1-\tan\alpha}{1+\tan\alpha}$ (3) $\frac{1+\sin2\alpha}{1-\sin2\alpha}$ (4) $\frac{1+\tan\alpha}{1-\tan\alpha}$

19. A particle is projected at an angle of 30° from horizontal at a speed of 60 m/s. The height traversed by the particle in the first second is $h_0$ and height traversed in the last second, before it reaches the maximum height, is $h_1$. The ratio $h_0 : h_1$ is________.
[Take, $g = 10\ m/s^2$] [22 Jan, 2025 (Shift-I)]

20. Two projectiles are fired with same initial speed from same point on ground at angles of $(45° - \alpha)$ and $(45° + \alpha)$, respectively, with the horizontal direction. The ratio of their maximum heights attained is: [29 Jan, 2025 (Shift-I)]
(a) $\frac{1-\tan\alpha}{1+\tan\alpha}$ (b) $\frac{1-\sin2\alpha}{1+\sin2\alpha}$ (c) $\frac{1+\sin\alpha}{1-\sin\alpha}$ (d) $\frac{1+\sin2\alpha}{1-\sin2\alpha}$

21. Projectiles $A$ and $B$ are thrown at angles of 45° and 60° with vertical respectively from top of a 400 m high tower. If their ranges and times of flight are same, the ratio of their speeds of projection $v_A : v_B$ is: [30 Jan, 2024 (Shift-II)]
(a) $1:\sqrt{3}$ (b) $\sqrt{2}:1$ (c) $1:2$ (d) $1:\sqrt{2}$

22. A particle moving in a circle of radius $R$ with uniform speed takes time $T$ to complete one revolution. If this particle is projected with the same speed at an angle $\theta$ to the horizontal, the maximum height attained by it is equal to $4R$. The angle of projection $\theta$ is then given by: [1 Feb, 2024 (Shift-I)]
(a) $\sin^{-1}\left[\frac{2gT^2}{\pi^2 R}\right]^{\frac{1}{2}}$ (b) $\sin^{-1}\left[\frac{\pi^2 R}{2gT^2}\right]^{\frac{1}{2}}$
(c) $\cos^{-1}\left[\frac{2gT^2}{\pi^2 R}\right]^{\frac{1}{2}}$ (d) $\cos^{-1}\left[\frac{\pi R}{2gT^2}\right]^{\frac{1}{2}}$

23. The maximum height reached by a projectile is 64 m. If the initial velocity is halved, the new maximum height of the projectile is ________ m. [05 April, 2024 (Shift-II)]

24. The angle of projection for a projectile to have same horizontal range and maximum height is: [08 April, 2024 (Shift-II)]
(a) $\tan^{-1}(2)$ (b) $\tan^{-1}(4)$ (c) $\tan^{-1}\left(\frac{1}{4}\right)$ (d) $\tan^{-1}\left(\frac{1}{2}\right)$

25. The range of the projectile projected at an angle of 15° with horizontal is 50 m. If the projectile is projected with same velocity at an angle of 45° with horizontal, then its range will be:
[10 April, 2023 (Shift-I)]
(a) $50\ m$ (b) $50\sqrt{2}\ m$ (c) $100\ m$ (d) $100\sqrt{2}\ m$

26. The initial speed of a projectile fired from ground is $u$. At the highest point during its motion, the speed of projectile is $\frac{\sqrt{3}}{2}u$. The time of flight of the projectile is: [31 Jan, 2023 (Shift-I)]
(a) $\frac{u}{2g}$ (b) $\frac{u}{g}$ (c) $\frac{2u}{g}$ (d) $\frac{\sqrt{3}u}{g}$

27. The trajectory of projectile, projected from the ground is given by $y = x - \frac{x^2}{20}$. Where $x$ and $y$ are measured in meter. The maximum height attained by the projectile will be. [8 April, 2023 (Shift-II)]
(a) $5\ m$ (b) $10\sqrt{2}\ m$ (c) $200\ m$ (d) $10\ m$

28. For a body projected at an angle with the horizontal from the ground, choose the correct statement. [1 Feb, 2023 (Shift-II)]
(a) Gravitational potential energy is maximum at the highest point.
(b) The horizontal component of velocity is zero at highest point.
(c) The vertical component of momentum is maximum at the highest point.
(d) The kinetic energy (K.E.) is zero at the highest point of projectile motion.

29. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R.**
**Assertion A:** When a body is projected at an angle 45°, it's range is maximum.
**Reason R:** For maximum range, the value of $\sin2\theta$ should be equal to one.
In the light of the above statements, choose the correct answer from the options given below: [6 April, 2023 (Shift-I)]
(a) Both A and R are correct but R is NOT the correct explanation of A
(b) Both A and R are correct R is the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

30. A projectile is projected at 30° from horizontal with initial velocity 40 $ms^{-1}$. The velocity of the projectile at $t = 2$ s from the start will be: (Given $g = 10\ m/s^2$) [11 April, 2023 (Shift-II)]

(a) $20\sqrt{3}\ ms^{-1}$ (b) $40\sqrt{3}\ ms^{-1}$
(c) 20 $ms^{-1}$ (d) Zero

31. Two objects are projected with the same velocity '$u$' however at different angles $\alpha$ and $\beta$ with the horizontal. If $\alpha + \beta = 90°$, the ratio of horizontal range of the first object to the 2nd object will be: [25 Jan, 2023 (Shift-II)]

(a) 4 : 1 (b) 2 : 1 (c) 1 : 2 (d) 1 : 1

32. Two projectiles are projected at 30° and 60° with the horizontal with the same speed. The ratio of the maximum height attained by the two projectiles respectively is: [10 April, 2023 (Shift-II)]

(a) $2:\sqrt{3}$ (b) $\sqrt{3}:1$ (c) 1 : 3 (d) $1:\sqrt{3}$

33. The maximum vertical height to which a man can throw a ball is 136 $m$. The maximum horizontal distance upto which he can throw the same ball is [24 Jan, 2023 (Shift-I)]

(a) 192 $m$ (b) 136 $m$ (c) 272 $m$ (d) 68 $m$

34. A projectile fired at 30° to the ground is observed to be at same height at time 3$s$ and 5s after projection, during its flight. The speed of projection of the projectile is __________ $ms^{-1}$ (Given $g = 10\ ms^{-2}$) [11 April, 2023 (Shift-I)]

35. Two bodies are projected from ground with same speeds 40$ms^{-1}$ at two different angles with respect to horizontal. The bodies were found to have same range. If one of the body was projected at an angle of 60°, with horizontal then sum of the maximum heights, attained by the two projectiles, is______ m. (Given $g = 10 ms^{-2}$) [31 Jan, 2023 (Shift-II)]

36. A stone is projected at angle 30° to the horizontal. The ratio of kinetic energy of the stone at point of projection to its kinetic energy at the highest point of flight will be [29 Jan, 2023 (Shift-I)]

(a) 1 : 2 (b) 1 : 4 (c) 4 : 1 (d) 4 : 3

37. Two projectiles $A$ and $B$ are thrown with initial velocities of 40 m/s and 60 m/s at angles 30° and 60° with the horizontal respectively. The ratio of their ranges respectively is ($g = 10\ m/s^2$) [8 April, 2023 (Shift-I)]

(a) $\sqrt{3}:2$ (b) $2:\sqrt{3}$ (c) 1 : 1 (d) 4 : 9

38. Given below are two statements. One is labelled as **Assertion A** and the other is labelled as **Reason R.**

**Assertion A:** Two identical balls $A$ and $B$ thrown with same velocity '$u$' at two different angles with horizontal attained the same range $R$. If $A$ and $B$ reached the maximum height $h_1$ and $h_2$ respectively, then $R = 4\sqrt{h_1 h_2}$

**Reason R:** Product of said heights.

$$h_1 h_2 = \left(\frac{u^2 \sin^2\theta}{2g}\right)\cdot\left(\frac{u^2\cos^2\theta}{2g}\right)$$

Choose the correct answer: [25 June, 2022 (Shift-II)]

(a) Both $A$ and $R$ are true and $R$ is the correct explanation of $A$.
(b) Both $A$ and $R$ are true but $R$ is NOT the correct explanation of $A$.
(c) $A$ is true but $R$ is false.
(d) $A$ is false but $R$ is true.

39. A projectile is launched at an angle '$\alpha$' with the horizontal with a velocity 20 $ms^{-1}$. After 10 s, it inclination with horizontal is '$\beta$'. The value of $\tan\beta$ will be: ($g = 10\ ms^{-2}$). [27 June, 2022 (Shift-I)]

(a) $\tan\alpha + 5\sec\alpha$ (b) $\tan\alpha - 5\sec\alpha$
(c) $2\tan\alpha - 5\sec\alpha$ (d) $2\tan\alpha + 5\sec\alpha$

40. Two projectiles are thrown with same initial velocity making an angle of 45° and 30° with the horizontal respectively. The ratio of their respective ranges will be: [26 July, 2022 (Shift-II)]

(a) $1:\sqrt{2}$ (b) $\sqrt{2}:1$ (c) $2:\sqrt{3}$ (d) $\sqrt{3}:2$

41. A body of mass 10 kg is projected at an angle of 45° with the horizontal. The trajectory of the body is observed to pass through a point (20, 10). If $T$ is the time of flight, then its momentum vector, at time $t = \frac{T}{\sqrt{2}}$ is______

[Take $g = 10\ m/s^2$] [27 July, 2022 (Shift-II)]

(a) $100\hat{i} + (100\sqrt{2} - 200)\hat{j}$ (b) $100\sqrt{2}\hat{i} + (100 - 200\sqrt{2})\hat{j}$
(c) $100\hat{i} + (100 - 200\sqrt{2})\hat{j}$ (d) $100\sqrt{2}\hat{i} + (100\sqrt{2} - 200)\hat{j}$

42. A projectile is projected with velocity of 25 m/s at an angle $\theta$ with the horizontal. After $t$ seconds its inclination with horizontal becomes zero. If $R$ represents horizontal range of the projectile, the value of $\theta$ will be: . [24 June, 2022 (Shift-I)]

(a) $\frac{1}{2}\sin^{-1}\left(\frac{5t^2}{4R}\right)$ (b) $\frac{1}{2}\sin^{-1}\left(\frac{4R}{5t^2}\right)$
(c) $\tan^{-1}\left(\frac{4t^2}{5R}\right)$ (d) $\cot^{-1}\left(\frac{R}{20t^2}\right)$

43. A ball is projected from the ground with a speed 15 $ms^{-1}$ at an angle $\theta$ with horizontal so that its range and maximum height are equal. Then '$\tan\theta$' will be equal to: [25 July, 2022 (Shift-II)]

(a) 1/4 (b) 1/2 (c) 2 (d) 4

44. A person can throw a ball upto a maximum range of 100 m. How high above the ground he can throw the same ball? [29 June, 2022 (Shift-II)]

(a) 25 m (b) 50 m (c) 100 m (d) 200 m

45. A body is projected from the ground at an angle of 45° with horizontal. Its velocity after 2s is 20 $ms^{-1}$. The maximum height reached by the body during its motion is ______m. (use $g = 10ms^{-2}$) [24 June, 2022 (Shift-II)]

46. If the initial velocity in horizontal direction of a projectile is unit vector $\hat{i}$ and the equation of trajectory is $y = 5x(1 - x)$. The $y$ component vector of the initial velocity is __________$\hat{j}$. (Take $g = 10\ m/s^2$) [26 July, 2022 (Shift-I)]

47. An object is projected in the air with initial velocity $u$ at an angle $\theta$. The projectile motion is such that the horizontal range $R$, is maximum. Another object is projected in the air with a horizontal range half of the range of first object. The initial velocity remains same in both the case. The value of the angle of projection, at which the second object is projected, will be ______________ degree. [29 July, 2022 (Shift-I)]

48. A ball is projected with kinetic energy $E$, at an angle of 60° to the horizontal. The kinetic energy of this ball at the highest point of its flight will become: [29 July, 2022 (Shift-I)]

(a) Zero (b) $\frac{E}{2}$ (c) $\frac{E}{4}$ (d) $E$

49. Two projectile thrown at 30° and 45° with the horizontal respectively, reach the maximum height in same time. The ratio of their initial velocities is [26 July, 2022 (Shift-I)]

(a) $1:\sqrt{2}$ (b) 2 : 1 (c) $\sqrt{2}:1$ (d) 1 : 2

**50.** A ball of mass m is thrown vertically upward. Another ball of mass 2 m is thrown an angle θ with the vertical. Both the balls stay in air for the same period of time. The ratio of the heights attained by the two balls respectively is $\frac{1}{x}$. The value of $x$ is ________. **[27 July, 2022 (Shift-I)]**

**51.** A body is projected at $t = 0$ with a velocity 10 $ms^{-1}$ at an angle of 60° with the horizontal. The radius of curvature of its trajectory at $t = 1$ s is $R$. Neglecting air resistance and taking acceleration due to gravity $g = 10$ $ms^{-2}$ the radius of $R$ is: **[11 Jan, 2019 (Shift-I)]**

(*a*) 10.3 m (*b*) 2.8 m (*c*) 2.5 m (*d*) 5.1 m

**52.** Two guns $A$ and $B$ can fire bullets at speed 1 km/s and 2 km/s respectively. From a point on a horizontal ground, they are fired in all possible directions. The ratio of maximum areas covered by the bullets fired by the two guns, on the ground is: **[10 Jan, 2019 (Shift-I)]**

(*a*) 1 : 16 (*b*) 1 : 2 (*c*) 1 : 4 (*d*) 1 : 8

**53.** A shell is fired from a fixed artillery gun with an initial speed $u$ such that it hits the target on the ground at a distance $R$ from it. If $t_1$ and $t_2$ are the values of the time taken by it to hit the target in two possible ways, the product $t_1t_2$ is: **[12 April, 2019 (Shift-I)]**

(*a*) $R/g$ (*b*) $R/4g$ (*c*) $2R/g$ (*d*) $R/2g$

**54.** Two particles are projected from the same point with the same speed u such that they have the same range $R$, but different maximum heights, $h_1$ and $h_2$. Which of the following is correct? **[12 April, 2019 (Shift-II)]**

(*a*) $R^2 = 2h_1 h_2$ (*b*) $R^2 = 16h_1 h_2$ (*c*) $R^2 = 4h_1 h_2$ (*d*) $R^2 = h_1 h_2$

**55.** The trajectory of a projectile near the surface of the earth is given as $y = 2x - 9x^2$. If it were launched at an angle $\theta_0$ with speed $v_0$ then ($g = 10$ $ms^{-2}$) **[12 April, 2019 (Shift-I)]**

(*a*) $\theta_0 = \cos^{-1}\left(\frac{2}{\sqrt{5}}\right)$ and $v_0 = \frac{3}{5}\,ms^{-1}$

(*b*) $\theta_0 = \sin^{-1}\left(\frac{1}{\sqrt{5}}\right)$ and $v_0 = \frac{5}{3}\,ms^{-1}$

(*c*) $\theta_0 = \sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$ and $v_0 = \frac{3}{5}\,ms^{-1}$

(*d*) $\theta_0 = \cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$ and $v_0 = \frac{5}{3}\,ms^{-1}$

## Horizontal Projectile

**56.** A helicopter flying horizontally with a speed of 360 km/h at an altitude of 2 km, drops an object at an instant. The object hits the ground at a point O, 20 s after it is dropped. Displacement of 'O' from the position of helicopter where the object was released is : (use acceleration due to gravity $g = 10$ m/s² and neglect air resistance) **[07 April, 2025 (Shift-II)]**

(*a*) $2\sqrt{5}$ km (*b*) 4 km (*c*) 7.2 km (*d*) $2\sqrt{2}$ km

**57.** A ball rolls off the top of a stairway with horizontal velocity u. The steps are 0.1 m high and 0.1 m wide. The minimum velocity u with which that ball just hits the step 5 of the stairway will be $\sqrt{x}$ $ms^{-1}$ where $x$ = ________ [use $g = 10$ m/s²]. **[29 Jan, 2024 (Shift-I)]**

**58.** A body of mass M thrown horizontally with velocity v from the top of the tower of height $H$ touches the ground at a distance of 100m from the foot of the tower. A body of mass 2 $M$ thrown at a velocity v/2 from the top of the tower of height 4$H$ will touch the ground at a distance of ......... m. **[08 April, 2024 (Shift-II)]**

**59.** A child stands on the edge of the cliff 10 $m$ above the ground and throws a stone horizontally with an initial speed of 5 $ms^{-1}$. Neglecting the air resistance, the speed with which the stone hits the ground will be $ms^{-1}$ (given, $g = 10$ $ms^{-1}$). **[1 Feb, 2023 (Shift-I)]**

(*a*) 20 (*b*) 15 (*c*) 30 (*d*) 25

**60.** A helicopter rises from rest on the ground vertically upwards with a constant acceleration g. A food packet is dropped from the helicopter when it is at a height $h$. The time taken by the packet to reach the ground is close to [$g$ is the acceleration due to gravity] **[5 Sep, 2020 (Shift-I)]**

(*a*) $t = 3.4\sqrt{\left(\frac{h}{g}\right)}$ (*b*) $t = 1.8\sqrt{\frac{h}{g}}$

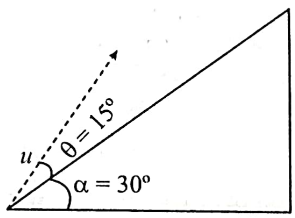
(*c*) $t = \sqrt{\frac{2h}{3g}}$ (*d*) $t = \frac{2}{3}\sqrt{\left(\frac{h}{g}\right)}$

## Projectile Motion on an Inclined Plane

**61.** A plane is inclined at an angle α = 30° with a respect to the horizontal. A particle is projected with a speed $u = 2$ $ms^{-1}$ from the base of the plane, making an angle θ = 15° with respect to the plane as shown in the figure.

The distance from the base, at which the particle hits the plane is close to: **[10 April, 2019 (Shift-II)]**

(Take $g = 10$ $ms^{-2}$)




(*a*) 14 cm (*b*) 20 cm (*c*) 18 cm (*d*) 26 cm

## Two Dimensional Relative Motion

**62.** A river is flowing from west to east direction with speed of 9 km $h^{-1}$. If a boat capable of moving at a maximum speed of 27 km $h^{-1}$ in still water, crosses the river in half a minute, while moving with maximum speed at an angle of 150° to direction of river flow, then the width of the river is: **[02 April, 2025 (Shift-I)]**

(*a*) 300 m (*b*) 112.5 m

(*c*) 75 m (*d*) $112.5\times\sqrt{3}$ m

**63.** The maximum speed of a boat in still water is 27 km/h. Now this boat is moving downstream in a river flowing at 9 km/h. A man in the boat throws a ball vertically upwards with speed of 10 m/s. Range of the ball as observed by an observer at rest on the river bank, is ________ cm. (Take $g = 10$ m/s²) **[29 Jan, 2025 (Shift-I)]**

**64.** The speed of a swimmer is 4 km $h^{-1}$ in still water. If the swimmer makes his strokes normal to the flow of river of width 1 km, he reaches a point 750 m down the stream on the opposite bank.

The speed of the river water is ________ km $h^{-1}$. **[31 Jan, 2023 (Shift-I)]**

**65.** A fighter jet is flying horizontally at a certain with a speed of 200 $ms^{-1}$. When it passes directly overhead an anti-aircraft gun, a bullet is fired from the gun, at an angle θ with the horizontal, to hit the jet. If the bullet speed is 400 m/s, the value of θ will be ________. **[26 June, 2022 (Shift-I)]**

66. A girl standing on road holds her umbrella at 45° with the vertical to keep the rain away. If she starts running without umbrella with a speed of $15\sqrt{2}$ kmh$^{-1}$, the rain drops hit her head vertically. The speed of rain drops with respect to the moving girl is:

[27 June, 2022 (Shift-I)]

(a) 30 kmh$^{-1}$ (b) $\frac{25}{\sqrt{2}}$ kmh$^{-1}$ (c) $\frac{30}{\sqrt{2}}$ kmh$^{-1}$ (d) 25 kmh$^{-1}$

67. When a car is at rest, its driver sees rain drops falling on it vertically. When driving the car with speed v, he sees that rain drops are coming at an angle 60° from the horizontal. On further increasing the speed of the car to $(1 + \beta)v$, this angle changes to 45°. The value of β is close to [6 Sep, 2022 (Shift-II)]

(a) 0.37 (b) 0.41 (c) 0.73 (d) 0.50

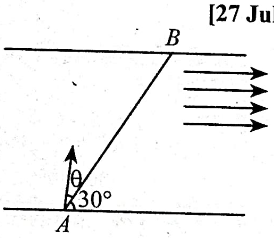
68. A butterfly is flying with a velocity $4\sqrt{2}$ m/s in North – East direction. Wind is slowly blowing at 1 m/s from North to South. The resultant displacement of the butterfly in 3 seconds is:

[20 July, 2021 (Shift-I)]

(a) 15m (b) 20m (c) 3 m (d) $12\sqrt{2}m$

69. A swimmer wants to cross a river from point $A$ to point $B$. Line $AB$ makes an angle of 30° with the flow of river. Magnitude of velocity of the swimmer is same as that of the river. The angle θ with the line $AB$ should be ________°, so that the swimmer reaches point $B$.

[27 July, 2021 (Shift-II)]




70. A swimmer can swim with velocity of 12 km/ h in still water. Water flowing in a river has velocity 6 km/ h. The direction with respect to the direction of flow of river water he should swim in order to reach the point on the other bank just opposite to his starting point is ______°. (Round off to the Nearest Integer)

(Find the angle in degrees) [16 March, 2021 (Shift-II)]

71. A person is swimming with a speed of 10 m/s at an angle of 120° with the flow and reaches to a point directly opposite on the other side of the river. The speed of the flow is '$x$' m/s. The value of '$x$' to the nearest integer is ________. [18 March, 2021 (Shift-I)]

72. Ship $A$ is sailing towards north-east with velocity $\vec{v} = 30\hat{i} + 50\hat{j}$ km/hr where $\hat{i}$ points east and $\hat{j}$, north. Ship $B$ is at a distance of 80 km east and 150 km north of Ship $A$ and is sailing towards west at 10 km/hr. $A$ will be at minimum distance from $B$ in:

[8 April, 2019 (Shift-I)]

(a) 4.2 hrs. (b) 2.2 hrs. (c) 3.2 hrs. (d) 2.6 hrs.

73. The stream of a river is flowing with a speed of 2km/h. A swimmer can swim at a speed of 4km/h. What should be the direction of the swimmer with respect to the flow of the river to cross the river straight? [9 April, 2019 (Shift-I)]

(a) 60° (b) 150° (c) 90° (d) 120°

## Uniform Circular Motion

74. If the radius of curvature of the path of two particles of same mass are in the ratio 3:4, then in order to have constant centripetal force, their velocities will be in the ratio of: [29 Jan, 2024 (Shift-I)]

(a) $\sqrt{3}:2$ (b) $1:\sqrt{3}$ (c) $\sqrt{3}:1$ (d) $2:\sqrt{3}$

75. A particle is moving in a circle of radius 50 cm in such a way that at any instant the normal and tangential components of its acceleration are equal. If its speed at $t = 0$ is 4 m/s, the time taken to complete the first revolution will be $\frac{1}{\alpha}[1-e^{-2\pi}]s$, where $\alpha =$ ________. is

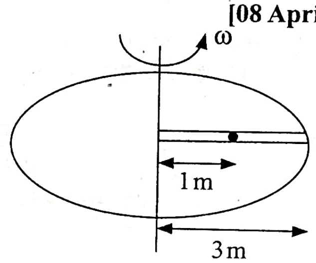
[29 Jan, 2024 (Shift-II)]

76. A clock has 75 cm, 60 cm long second hand and minute hand respectively. In 30 minutes duration the tip of second hand will travel $x$ distance more than the tip of minute hand. The value of $x$ in meter is nearly (Take $\pi = 3.14$): [08 April, 2024 (Shift-I)]

(a) 139.4 (b) 140.5 (c) 220.0 (d) 118.9

77. A circular table is rotating with an angular velocity of ω rad/s about its axis (see figure). There is a smooth groove along a radial direction on the table. A steel ball is gently placed at a distance of 1 m on the groove. All the surface are smooth. If the radius of the table is 3 m, the radial velocity of the ball w.r.t. the table at the time ball leaves the table is $x\sqrt{2}\omega$ m/s, where the value of $x$ is..........

[08 April, 2024 (Shift-II)]




78. An object moves at a constant speed along a circular path in a horizontal plane with centre at the origin. When the object is at $x = +2m$, its velocity is $-4\hat{j}m/s$. The object's velocity ($v$) and acceleration ($a$) at $x = -2m$ will be [29 Jan, 2023 (Shift-II)]

(a) $v = 4\hat{i}m/s, a = 8\hat{j}m/s^2$ (b) $v = 4\hat{j}m/s, a = 8\hat{i}m/s^2$

(c) $v = -4\hat{j}m/s, a = 8\hat{i}m/s^2$ (d) $v = -4\hat{i}m/s, a = -8\hat{j}m/s^2$

79. A car is moving on a circular path of radius 600 m such that the magnitudes of the tangential and centripetal acceleration are equal. Time taken by the car to complete first quarter of revolution, if it is moving with an initial speed of 54 km/hr is $t(1-e^{-\pi/2})s$. The value of $t$ is ________.

[29 Jan, 2023 (Shift-II)]

80. A stone tied to 180 cm long string at its end is making 28 revolutions in horizontal circle in every minute. The magnitude of acceleration of stone is $\frac{1936}{x}$ ms$^{-2}$. The value of $x$ ____________.

$\left(\text{Take } \pi = \frac{22}{7}\right)$ [30 Jan, 2023 (Shift-II)]

81. A ball is spun with angular acceleration $\alpha = 6t^2 - 2t$ where $t$ is in second and α is in rads$^{-1}$. At $t = 0$, the ball has angular velocity of 10 rads$^{-1}$ and angular position of 4 rad. The most appropriate expression for the angular position of the ball is: [28 June, 2022 (Shift-II)]

(a) $\frac{3}{2}t^4 - t^2 + 10t$ (b) $\frac{t^4}{2} - \frac{t^3}{3} + 10t + 4$

(c) $\frac{2t^4}{3} - \frac{t^3}{6} + 10t + 12$ (d) $2t^4 - \frac{t^3}{2} + 5t + 4$

82. Motion of a particle in $x - y$ plane is described by a set of following equations $x = 4\sin\left(\frac{\pi}{2} - \omega t\right)m$ and $y = 4\sin(\omega t)$m. The path of the particle will be: [28 June, 2022 (Shift-I)]

(a) Circular (b) Helical (c) Parabolic (d) Elliptical

**83.** A fly wheel is accelerated uniformly from rest and rotates through 5 rad in the first second. The angle rotated by the fly wheel in the next second, will be: **[24 June, 2022 (Shift-II)]**

(*a*) 7.5 rad (*b*) 15 rad (*c*) 20 rad (*d*) 30 rad

**84.** A body rotating with an angular speed of 600rpm is uniformly accelerated to 1800rpm in 10 sec. The number of rotations made in the process is ________. **[20 July, 2021 (Shift-II)]**

**85.** A particle moves such that its position vector $\vec{r}(t) = \cos \omega t\ \hat{i} + \sin \omega t\ \hat{j}$ where $\omega$ is a constant and $t$ is time. Then which of the following statements is true for the velocity $\vec{v}(t)$ and acceleration $\vec{a}(t)$ of the particle: **[8 Jan, 2020 (Shift-II)]**

(*a*) $\vec{v}$ is perpendicular to $\vec{r}$ and $\vec{a}$ is directed towards the origin.

(*b*) $\vec{v}$ and $\vec{a}$ both are parallel to $\vec{r}$.

(*c*) $\vec{v}$ is perpendicular to $\vec{r}$ and $\vec{a}$ is directed away from the origin.

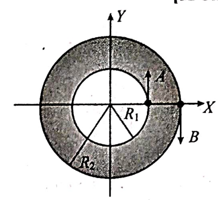
(*d*) $\vec{v}$ and $\vec{a}$ both are perpendicular to $\vec{r}$.

**86.** A particle is moving along a circular path with a constant speed of 10 ms$^{-1}$. What is the magnitude of the change in velocity of the particle, when it moves through an angle of 60° around the centre of the circle? **[11 Jan, 2019 (Shift-I)]**

(*a*) $10\sqrt{3}$ m/s (*b*) 0

(*c*) $10\sqrt{2}$ m/s (*d*) 10 m/s

**87.** Two particles $A$, $B$ are moving on two concentric circles of radii $R_1$ and $R_2$ with equal angular speed $\omega$. At $t = 0$, their positions and direction of motion are shown in the figure: **[12 Jan, 2019 (Shift-II)]**




The relative velocity $\vec{v_A} - \vec{v_B}$ at $t = \dfrac{\pi}{2\omega}$ is given by:

(*a*) $\omega(R_1 + R_2)\hat{i}$ (*b*) $-\omega(R_1 + R_2)\hat{i}$

(*c*) $\omega(R_2 - R_1)\hat{i}$ (*d*) $\omega(R_1 - R_2)\hat{i}$

## ANSWER KEY

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1. (*d*) | 2. (*d*) | 3. [673] | 4. (*d*) | 5. [2] | 6. (*b*) | 7. [580] | 8. (*d*) | 9. (*d*) | 10. (*a*) |
| 11. (*c*) | 12. (*b*) | 13. (*a*) | 14. (*d*) | 15. (*d*) | 16. (*c*) | 17. (*a*) | 18. (*d*) | 19. [5] | 20. (*b*) |
| 21. (Bonus) | 22. (*a*) | 23. [16] | 24. (*b*) | 25. (*c*) | 26. (*b*) | 27. (*a*) | 28. (*a*) | 29. (*b*) | 30. (*a*) |
| 31. (*d*) | 32. (*c*) | 33. (*c*) | 34. [80] | 35. [80] | 36. (*d*) | 37. (*d*) | 38. (*a*) | 39. (*b*) | 40. (*c*) |
| 41. (*d*) | 42. (*d*) | 43. (*d*) | 44. (*b*) | 45. [20] | 46. [5] | 47. [15or75] | 48. (*c*) | 49. (*c*) | 50. [1] |
| 51. (*b*) | 52. (*a*) | 53. (*c*) | 54. (*b*) | 55. (*d*) | 56. (*d*) | 57. [2] | 58. [100] | 59. (*b*) | 60. (*a*) |
| 61. (*b*) | 62. (*b*) | 63. [2000] | 64. [3] | 65. [60] | 66. (*c*) | 67. (*c*) | 68. (*a*) | 69. [30] | 70. [120] |
| 71. [5] | 72. (*d*) | 73. (*d*) | 74. (*a*) | 75. [8] | 76. (*a*) | 77. [2] | 78. (*b*) | 79. [40] | 80. [125] |
| 81. (*b*) | 82. (*a*) | 83. (*b*) | 84. [200] | 85. (*a*) | 86. (*d*) | 87. (*c*) | | | |